Invisibility Cloak for Depth Deception

Adversarial patches against monocular depth estimation: from digital training to the physical channel.

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Outline

Task 1 -- Baseline

Fine-tune Depth-Anything-V2-Small on NYU Depth v2.

RMSE 0.349 m
δ1 93.5 %
δ2 98.97 %

Task 2 -- Attack

Train a 192^2 printable patch that drives the covered region to 10 m. Three versions, monotonic gain at every step.

v1 25.8 % > 8 m
v2 31.1 % > 8 m
v3a 53.0 % > 8 m

Task 3 -- Characterisation

Sweep digital, physical-simulation, and real-world conditions; identify what works, identify what breaks.

scene-agnostic across 6 NYU categories; gap to real-world quantified

Patch methodology adapted from Wu et al., "Making an Invisibility Cloak", ECCV 2020.

Task 1: Problem Statement: Why attack a depth model?

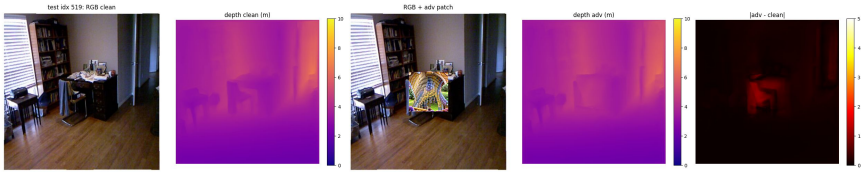
Monocular depth estimators are entering production in robotics, AR, and driver assistance. Unlike classifiers, their output is a dense metric field, so an attack succeeds only if it forces specific numeric depth values at specific spatial locations under messy imaging conditions.

Brown et al. (2017) and Wu et al. (2020) showed that a printable RGB patch can reliably fool image classifiers and hide a person from a detector.

The objective of this project was the creation of an attack pattern that could fool the depth estimator, making it detect huge distances in proximity with our patch.

We tried to adapt the patch to be working in different lighting scenarios and digital first.

Target model is Depth-Anything-V2-Small (indoor metric). Target depth $\tau = 10$ m. Patch is 192 × 192, RGB-constrained, TV-regularised so the optimised pattern stays printable.



v3a patch composited into an NYU scene. Right panel: predicted depth -- the patch region reads as far when the underlying surface is near.

Task 1: State of the Art

Monocular depth estimation

Depth-Anything V2 (Yang et al., 2024) is a transformer-based monocular depth predictor trained on ≈62 M images via teacher-student distillation. The metric-indoor variant fine-tuned here is the Small (≈25 M parameter) checkpoint, which reaches $\delta 1 \approx 0.93$ on NYU-v2 out of the box.

I adopt the SILog loss (Eigen et al., 2014) for fine-tuning, with a low backbone learning rate and a higher head learning rate to preserve pre-trained representations.

The entire solution was trained on top of a RTX4060Ti with 32B of RAM.

Physical-world adversarial patches

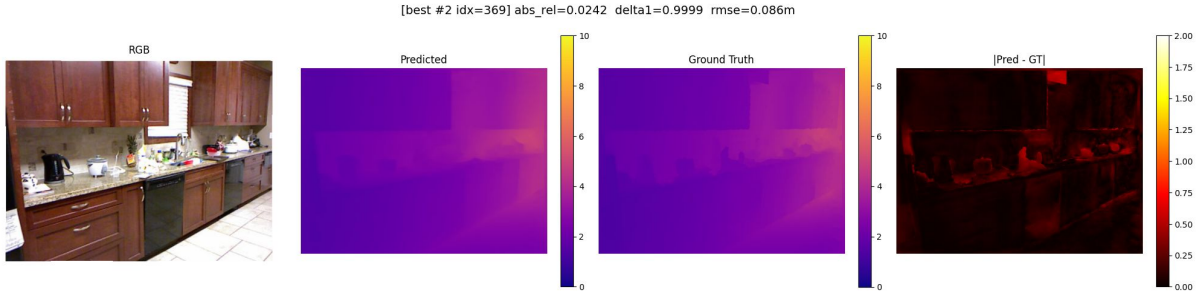
Brown et al. (2017) introduced the rectangular adversarial patch, trained with Expectation over Transformation. Thys et al. (2019) and Wu et al. (ECCV 2020) extended EoT for person-hiding patches against detectors, adding printability constraints, scale / rotation / affine augmentation, and outdoor physical validation.

Wu et al.'s EoT stack -- affine + colour jitter + blur / noise -- is my starting point. I add homography (pitch / yaw ±20°), display + moire + camera proxies for the screen-displayed deployment used in Task 3, and a wider in-plane rotation envelope (±30°).

Task 1: Dataset and evaluation protocol

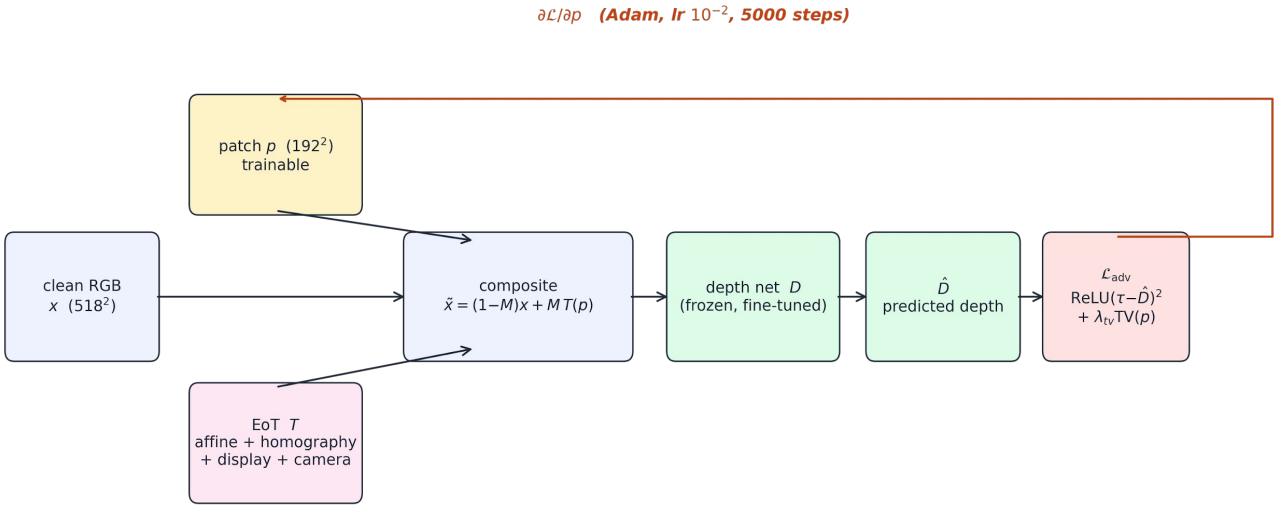
Split	Images	Use
NYU-v2 train (subset)	15 000	Task 1 fine-tuning -- 3 epochs, batch 6
NYU-v2 test (Eigen)	654	Baseline metrics, Task 3 sweeps
First 150 test × 3 seeds	450	Patch evaluation per condition
Own tablet captures	8	Real-world evaluation of v3a_physical

Eigen crop (45, 471, 41, 601) applied for all metrics. min_depth = 0.001 m, max_depth = 10.0 m. All patch numbers are in-mask: 'frac > 8 m' is over patch pixels, not the whole frame.



Five NYU-v2 test scenes with ground-truth depth and the fine-tuned baseline prediction. The sweeps in Task 3 cover bedroom, living room, kitchen, bathroom, office and dining-room categories.

Task 2: Method Proposed



$$\mathcal{L}_{\text{adv}}(p) \;=\; \frac{1}{|\mathcal{M}|} \sum_{(i,j)\,\in\,\mathcal{M}} \text{ReLU}\big(\tau - \hat{D}_{i,j}(x + T(p))\big)^2 \;+\; \lambda_{\text{tv}}\,\text{TV}(p), \quad \tau = 10\ \text{m}$$

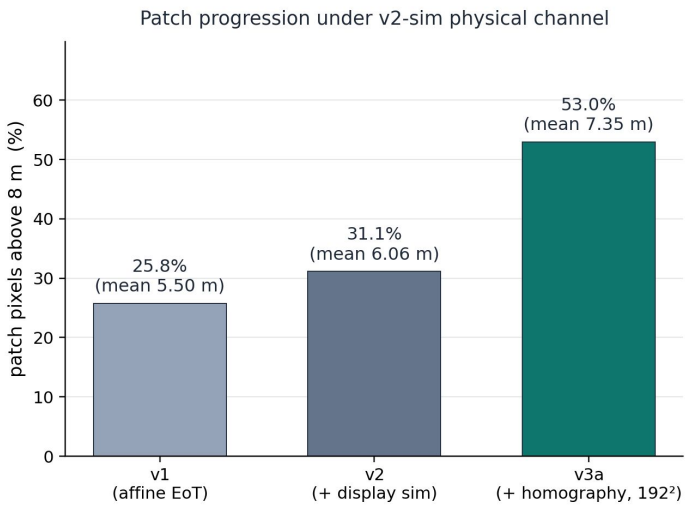
5000 steps | Adam, lr 1×10^{-2} | batch 4 | seed 42 | $\lambda_tv = 10^{-3}$ | patch 192^2 | trained on RTX 4060 Ti in 8.6 min (start mean depth in mask 2.69 m ⟶ final 6.38 m)

Task 2: Patch Description and Versions

Each row adds something: a bigger patch, a harder EoT stack at training time, or a harder simulator at evaluation. The bold row is the headline result.

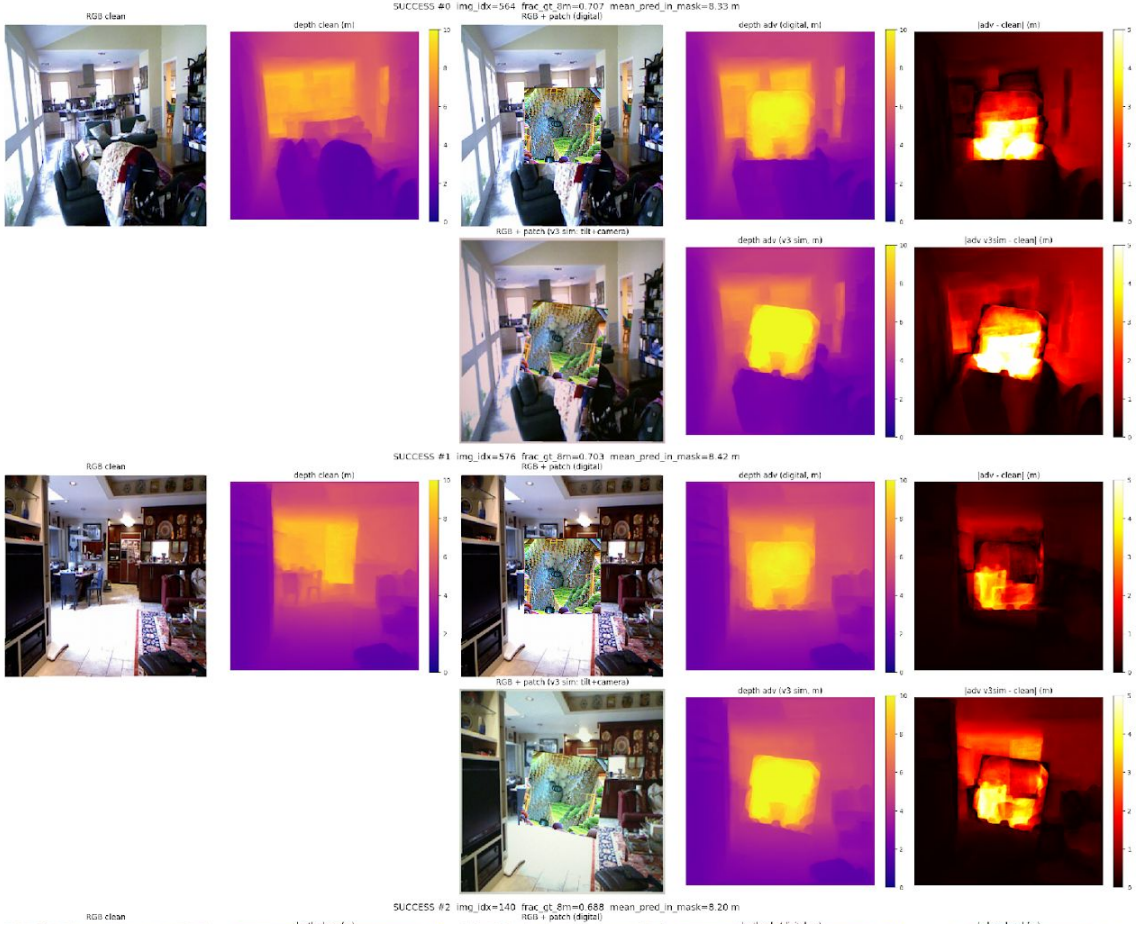
Patch	Size	EoT in training	Eval channel	mean (m)	> 8 m
v1	128^2	affine only (rot ±15°, scale 0.7-1.3)	v2-sim	5.50	25.8 %
v2	128^2	+ display + moire + camera + blur	v2-sim	6.06	31.1 %
v3a	192^2	+ homography (±20° pitch/yaw) + rot ±30°	digital	6.94	41.4 %
v3a	**192^2**	**(as above)**	**v2-sim**	**7.35**	**53.0 %**
v3a	192^2	(as above)	v3-sim	7.12	50.5 %

All evaluations: 150 NYU images × 3 seeds = 450 patch placements per row. Target depth τ = 10 m. Unattacked patch-region depth ≈ 1.5 m.

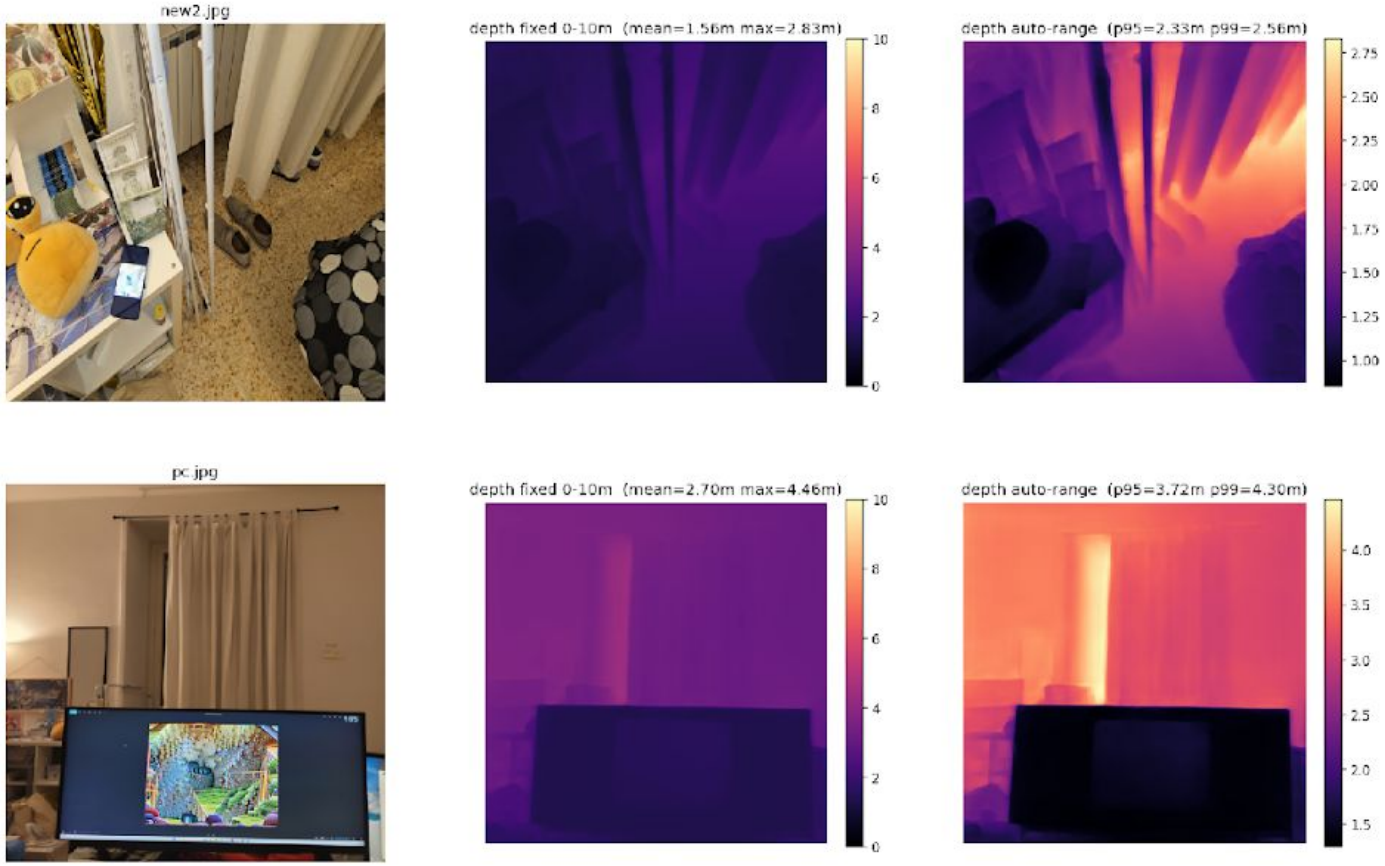


Monotonic gain v1 ⟶ v2 ⟶ v3a under the same v2-sim channel.

Task 3: Experiments Results (1/2)



Task 3: Experiments Results (2/2)



Task 3: What worked, what I characterised, what comes next

validated.

The patch formulation from Wu et al., adapted for metric depth with a ReLU-squared loss on the gap to 10 m, produces improvements from v1 (25.8 % of patch pixels above 8 m) through v2 (31.1 %) to v3a (53.0 %) as EoT realism and patch size grow. Per-scene results show the effect is scene-agnostic across six NYU indoor categories.

characterised.

The attack degrades gracefully and predictably under rotation, scale, brightness and contrast sweeps, and collapses exactly at the boundary of the EoT training envelope (rotation ±30°). On 8 tablet captures of the v3a_physical patch shown on a monitor, patch-region depth rose from a ≈1.5 m unattacked baseline to an aggregate mean of 2.00 m (max 3.65 m); best scene pc.jpg reached 2.70 m mean (max 4.46 m) -- a 30 to 80 % elevation, below the 10 m target but well above scene noise.

next.

The delta between sim mean 7.35 m and real-world mean 2.00 m is consistent with a channel that the current EoT stack does not model: display-side subpixel structure and gamma, plus camera-side demosaic, JPEG and white-balance applied on top of a printed patch. The natural follow-up is a printed physical patch photographed directly -- removing one compression stage and letting the EoT train against the full physical channel.

References

[1] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, H. Zhao. Depth Anything V2. arXiv:2406.09414, 2024.

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[8] D. Eigen, C. Puhrsch, R. Fergus. Depth Map Prediction from a Single Image using a Multi-Scale Deep Network. NeurIPS, 2014. (SILog loss)